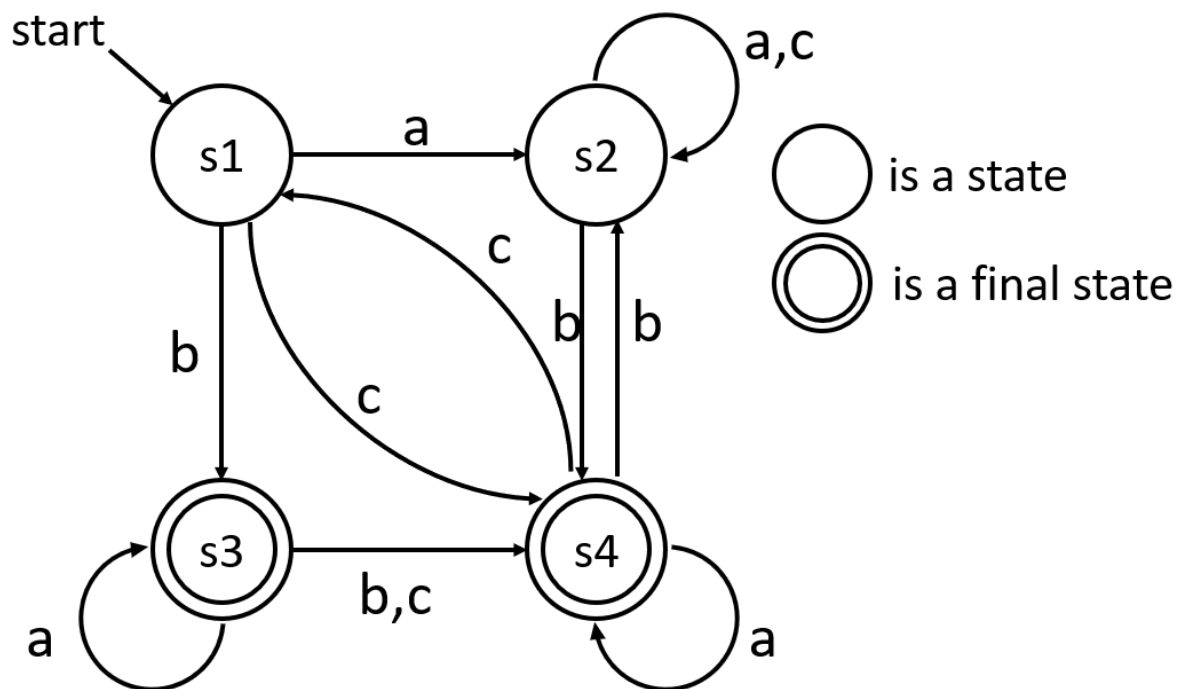


Problem B: Find the state

Subject

Description

Please refer to the image of DFA below and design a program that can process the string.



Input Format

A string of length N , consisting only of $\{a, b, c\}$ and ending with $\$$, where $1 \leq N \leq 20$.

Output Format

Output whether the automaton accepts the input.

If yes, output YES and the state where it stops, separated by a space.

If not, output NO.

No newline at the end.

Test Case

- Each test case input will consist of **only one line**.

input:
aaabc\$

output: NO
input: aabcb\$ output: YES s3
input: baac\$ output: YES s4
input: cccca\$ output: NO